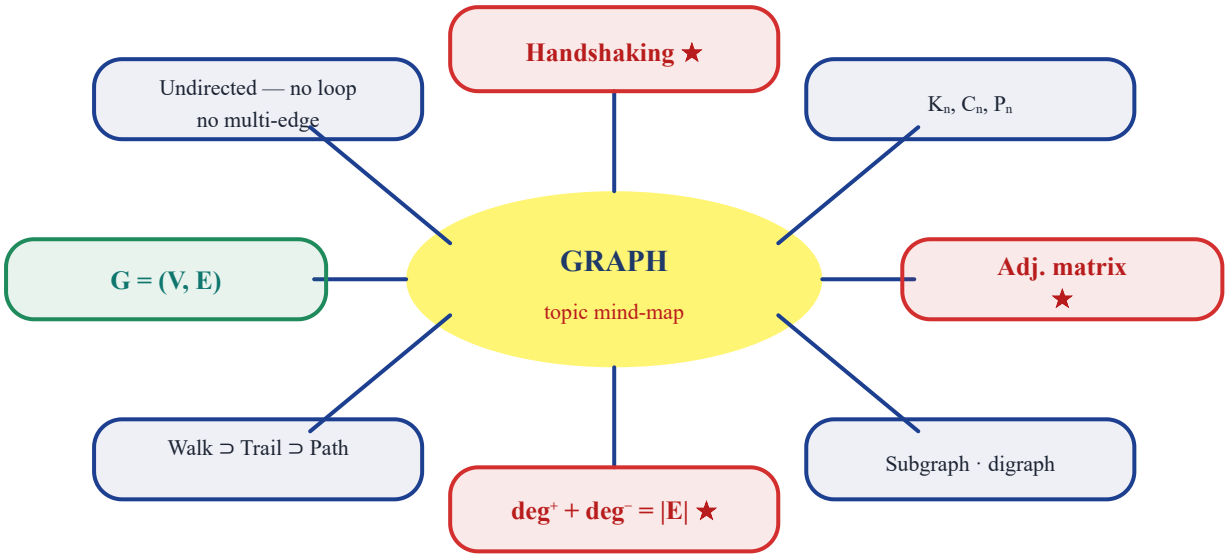
Graph Fundamentals

G = (V,E) · simple vs directed · degree · walk/path/cycle · K_n/C_n/$K_{m,n}$: exam-ready in 10 sheets



↑ the whole topic on one page: red = high-yield, green = the one definition every question builds from

Syllabus coverage

The syllabus line is "**Graph theory: basic definitions — simple graph, directed graph, subgraph, degree, walk, path, cycle, special graphs (K_n, C_n, $K_{m,n}$), representations (adjacency matrix, adjacency list)**". This lesson covers the foundation that Trees, Euler & Hamiltonian, and Planar Graphs lessons build on.

High-yield list

① G=(V,E) def
② no loop, no multi-edge
③ $deg^+ + deg^- = |E|$
④ Handshaking: $\Sigma deg = 2|E|$
⑤ Walk ⊇ Trail ⊇ Path ⊇ Cycle
⑥ K_n has $n(n-1)/2$ edges
⑦ Adj. matrix $O(V^2)$
⑧ Cycle C_n has n edges

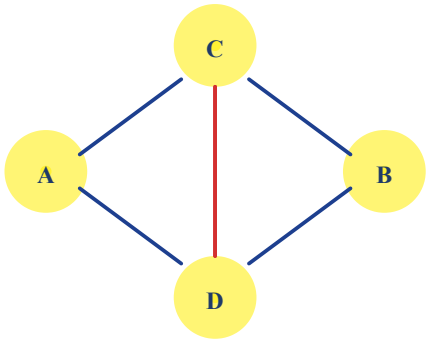
Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

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✓ What is a graph? Simple vs directed … p. 2
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1 What Is a Graph?

Definition: A **graph** G is an ordered pair **G = (V, E)**, where **V** is a non-empty set of **vertices** (nodes) and **E** is a set of **edges** (links). In an **undirected** graph each edge is an unordered pair {u, v}; in a **directed graph** (digraph) each edge is an ordered pair (u, v), written u → v.



left: undirected {A,B} … right: directed A→B

undirected: edge {A,B} same as {B,A} · directed: edge (A,B) ≠ (B,A)

Simple graph: An undirected graph with **no loops** (an edge from a vertex to itself) and **no multiple edges** between the same pair of vertices. Every pair of distinct vertices is joined by **at most one** edge.

Exam trap: (i) A **loop** at vertex v is an edge {v, v} — it breaks simplicity. (ii) **Multiple edges** between u and v also break simplicity. A graph with either is a **multigraph**, not a simple graph. (iii) "Simple" does **not** mean "small" — K_{100} is still a simple graph.

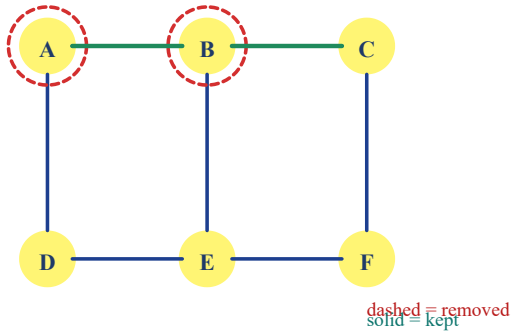
Term	Undirected	Directed (digraph)
Edge	{u, v} = {v, u}	(u, v) ≠ (v, u) in general
Loop	{v, v} — forbidden in simple graphs	(v, v) — self-loop
Multiple edge	Two edges {u,v} and {u,v} — forbidden	Two edges (u,v) and (u,v) — forbidden in simple digraph

Memory trick: "**Simple = clean.**" No self-loops, no double edges — just pure connections between distinct vertices. The moment you allow loops or parallel edges, you leave the world of simple graphs and enter **multigraphs**.

Worked example ✍ is this simple?: V= {A,B,C}, E={{A,B},{B,C},{C,A},{A,A}}. The edge {A,A} is a **loop** — so **not** simple. Remove it and the remaining graph **is** simple (it is the triangle C_3).

2 Subgraphs & Vertex Degrees

Definition: A **subgraph** H of G = (V, E) is a graph H = (V', E') such that **V' ⊆ V**, **E' ⊆ E**, and every edge in E' has both endpoints in V'. A **spanning subgraph** is a subgraph with V' = V (it keeps all vertices but may drop some edges).



H keeps A–B and C–D (renamed) edges; D and F are isolated in H

Concept	Undirected	Directed
Degree deg(v)	# of edges incident to v	—
In-degree $\deg^-(v)$	—	# of edges entering v
Out-degree $\deg^+(v)$	—	# of edges leaving v

Handshaking Lemma — undirected: $\Sigma \deg(v) = 2|E|$. Every edge touches exactly two ends, so it is counted twice in the degree sum.
Corollary: the number of odd-degree vertices is always even.

Handshaking Lemma — directed: $\Sigma \deg^+(v) = \Sigma \deg^-(v) = |E|$. Every directed edge leaves one vertex and enters another, so total out-degrees = total in-degrees = number of edges.

Memory trick: **"Every edge has two hands to shake."** Count every handshake from each vertex's perspective — each edge is shaken at both ends, so it contributes 2 to the total. Same idea for digraphs: each edge shakes once from the tail and once at the head.

Exam trap: (i) Handshaking holds for **multigraphs** too — the "simple" condition is not needed for the lemma. (ii) A loop in an undirected graph contributes **2** to the degree of its vertex (it touches the vertex at two ends). (iii) $\Sigma\deg^+ = \Sigma\deg^- = |E|$ for digraphs; do not multiply by 2 here.

Worked example ✍️ applying Handshaking: A simple undirected graph has 12 edges. By the lemma, $\Sigma \deg(v) = 2 \times 12 = \mathbf{24}$. If the graph has 6 vertices, the average degree is 24/6 = 4. Can every vertex have degree 4? Yes — that would be a 4-regular graph on 6 vertices (e.g. the complement of a perfect matching).

3 Walks, Trails, Paths & Cycles

Definition: Given a graph G, a **walk** of length k is a sequence $v_0, e_1, v_1, e_2, \ldots, e_k, v_k$ where each e_i joins v_{i-1} and v_i. Vertices and edges **may repeat**. A **trail** is a walk with **no repeated edge** (vertices may repeat). A **path** (or simple path) is a walk with **no repeated vertex** — hence no repeated edge automatically. A **cycle** is a path that starts and ends at the same vertex; for a simple graph the length must be ≥ 3.

Type	Vertices repeat?	Edges repeat?	Closed?
Walk	yes	yes	maybe
Trail	yes	no	maybe
Path	no	no	no (unless trivial)
Cycle	only start=end	no	yes

Inclusion diagram:

Walk — least restrictive

↓ no repeated edge

Trail

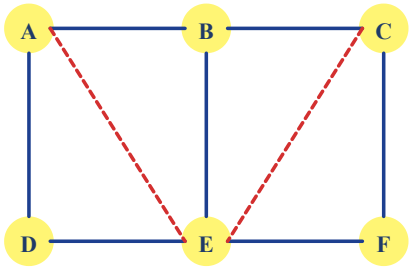
↓ no repeated vertex

Path

↓ closed

Cycle — most restrictive

Every path is a trail is a walk. Not every walk is a trail; not every trail is a path.



A→B→E→D→B→C is a walk but not a path (B repeats)

$v_0e_1v_1e_2\ldots e_kv_k$ — repeat freedom decreases as you move down the hierarchy

Exam trap: (i) "Path" in everyday language means any route; in graph theory it means **no repeated vertex**. (ii) A cycle must have **length ≥ 3** in a simple graph — two vertices with two edges between them (a multigraph) would be a cycle of length 2, but that is impossible in a simple graph. (iii) A single vertex with a self-loop is **not** a cycle in a simple graph (loops are forbidden).

Memory trick: "**No vertex repeated = path; no edge repeated = trail; both allowed = walk; closed path = cycle.**" Remember: path is the strictest common type; cycle is a path that returns home.

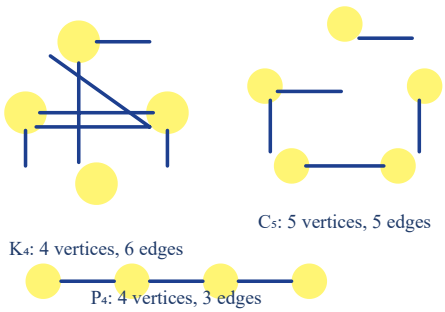
4 Special Graphs I: Complete, Cycle, Path

Graph	Structure	|E|	Degree of each vertex
K_n	every pair joined	$n(n-1)/2$	$n-1$
C_n	n vertices in one cycle	n	2
P_n	n vertices in a line	$n-1$	end = 1, internal = 2

Conditions: K_n requires $n \geq 1$ (K_1 has 0 edges). C_n requires $n \geq 3$ (a 2-cycle needs parallel edges, forbidden in simple graphs). P_n requires $n \geq 1$ (P_1 is a single vertex).

k-Regular graph: A graph where **every** vertex has the same degree k. K_n is $(n-1)$-regular. C_n is 2-regular. $K_{n,n}$ is n-regular. For a k-regular graph on n vertices, $|E| = nk/2$, so **nk must be even**.

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K_4 is 3-regular; C_5 is 2-regular; P_4 has two endpoints of degree 1

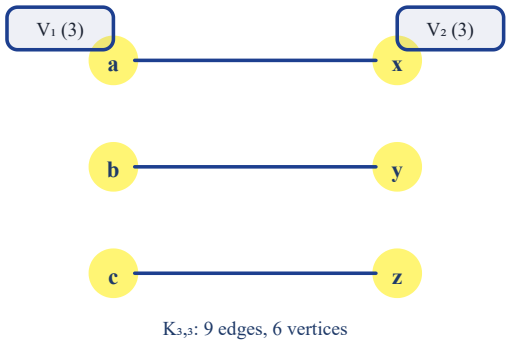
Exam trap: (i) K_1 has 0 edges, K_2 has 1 edge. K_3 is the triangle C_3. (ii) C_n exists only for $n \geq 3$ — C_1 and C_2 are not simple graphs. (iii) P_1 is a single vertex (0 edges); P_2 is a single edge. (iv) **K_n is (n−1)-regular** — every vertex has the same degree.

Memory trick: **"K gets all connections, C makes a ring, P draws a line."** K_n: everyone shakes hands with everyone. C_n: stand in a circle holding hands. P_n: stand in a queue.

5 Bipartite & Regular Graphs

Definition: A **bipartite graph** has its vertices partitioned into two sets V_1 and V_2 such that every edge joins a vertex in V_1 to one in V_2 — no edge joins two vertices within the same set. A **complete bipartite graph** $K_{m,n}$ joins **every** vertex of V_1 to **every** vertex of V_2.

Property	$K_{m,n}$
$\|V\|$	$m + n$
$\|E\|$	**$m \times n$**
Degree of $v \in V_1$	n
Degree of $v \in V_2$	m
Regular?	Only when $m = n$ (then n-regular)



$K_{3,3}$: 9 edges, 6 vertices

every vertex on the left connects to every vertex on the right

k-Regular graph: A graph where **every** vertex has the same degree k. K_n is (n−1)-regular. C_n is 2-regular. $K_{n,n}$ is n-regular. For a k-regular graph on n vertices, $|E| = nk/2$, so **nk must be even**.

König's theorem: A graph is **bipartite if and only if it has no odd cycle**. Equivalently: G is bipartite ⟺ every cycle in G has even length. **Odd cycle present ⟹ not bipartite; all cycles even ⟹ bipartite.**

Exam trap: (i) C_4 (square) **is** bipartite — vertices alternate between the two sets. C_5 (pentagon) is **not** bipartite — going around forces one vertex into both sets. (ii) $K_{2,3}$ is bipartite but **not** regular (degrees are 3 and 2). (iii) K_n is bipartite **only** for $n \leq 2$ — K_3 already contains a 3-cycle.

Memory trick: "**Even cycle = bipartite; odd cycle = not.**" Check any cycle: if all are even-length, the graph is bipartite. One odd cycle destroys bipartiteness entirely.

6 Graph Representation

Definition: A graph must be stored in some data structure. Three standard representations trade off space, lookup speed, and ease of traversal. The choice depends on whether the graph is **dense** ($|E| \approx |V|^2$) or **sparse** ($|E| \ll |V|^2$).

Representation	Space	Edge lookup	Iterate neighbours	Best for
Adjacency matrix	$O(V^2)$	$O(1)$	$O(V)$	dense graphs
Adjacency list	$O(V + E)$	$O(\deg(v))$	$O(\deg(v))$	sparse graphs
Edge list	$O(E)$	$O(E)$	$O(E)$	algorithm input

Worked example ✍️ a 4-vertex graph: G: V={1,2,3,4}, E={{1,2},{1,3},{2,3},{2,4}}.

Adjacency matrix (rows=cols=1,2,3,4):

```
0 1 1 0
1 0 1 1
1 1 0 0
0 1 0 0
```

Adjacency list:

```
1 → [2, 3]
2 → [1, 3, 4]
3 → [1, 2]
4 → [2]
```

Edge list:

```
(1,2) (1,3) (2,3) (2,4)
```

Exam trap: (i) For a **directed** graph, the adjacency matrix need **not** be symmetric: M[i][j]=1 means i→j, not j→i. (ii) Adjacency matrix uses V^2 space **even for sparse graphs** — avoid for large sparse graphs. (iii) Self-loops: matrix diagonal entry = 1 (undirected) or 1 per loop direction (directed). (iv) Multiple edges: adjacency matrix cannot represent them — use adjacency list with counts.

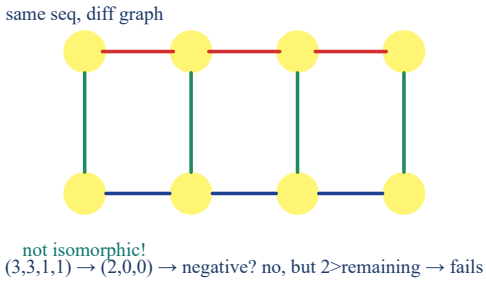
Memory trick: **"Matrix is square, list is lean."** $O(V^2)$ space is the matrix's liability; $O(V+E)$ is the list's strength. For dense graphs the difference shrinks; for sparse graphs the list wins hands down.

7 Degree Sequences & Handshaking Applications

Definition: The **degree sequence** of a graph is the list of vertex degrees written in non-increasing order, e.g. (3, 3, 2, 2, 1, 1). A sequence that can be realised as the degree sequence of **some** simple graph is called **graphical**.

Havel-Hakimi algorithm: To test whether a sequence $d = (d_1, d_2, \ldots, d_n)$ is graphical:

1. Sort in non-increasing order.
2. Remove d_1 and subtract 1 from the next d_1 terms.
3. Re-sort and repeat.
4. If you ever get a negative → **not graphical**. If you reach all zeros → **graphical**.



same degree sequence does not imply isomorphic graphs

Worked example ✍️ Havel-Hakimi on (3, 3, 1, 1):

1. Remove first 3, subtract 1 from next 3: (3,1,1) → (2, 0, 0).
2. Remove first 2, subtract 1 from next 2: (0,0) → (-1, -1). **Negative → not graphical.**

In a simple graph on 4 vertices, a vertex of degree 3 is adjacent to **all** others. Two such vertices force every other vertex to have degree ≥ 2, contradicting the 1s.

Corollaries of Handshaking: (i) $\Sigma \deg(v) = 2|E|$, so the sum is always **even**. (ii) The number of odd-degree vertices is **always even**. (iii) A regular graph of degree k on n vertices has $|E| = nk/2$, so **nk must be even**.

Exam trap: (i) **Same degree sequence ≠ isomorphic.** Two graphs can share the same degree sequence yet have different structures. (ii) Havel-Hakimi tests for **simple** graphs only — it does not apply to multigraphs. (iii) A sequence with an odd sum can **never** be graphical (violates Handshaking).

IT link: Degree-sequence analysis is used in network science to characterise real-world networks (social, biological, infrastructural) and compare them against random-graph models.

Night-Before-the-Exam Sheet

Fact sheet:

G = (V, E): vertices & edges	Simple: no loop, no multi-edge
Digraph: edges are ordered pairs	Subgraph: V'⊆V, E'⊆E, endpoints in V'
deg(v): # incident edges	$\deg^+(v)$: out-degree · $\deg^-(v)$: in-degree
Handshaking: $\Sigma \deg(v) = 2\|E\|$	Digraph: $\Sigma \deg^+ = \Sigma \deg^- = \|E\|$
Walk ⊃ Trail ⊃ Path ⊃ Cycle	K_n: $\|E\| = n(n-1)/2$, deg = n−1
C_n: $\|E\| = n$, 2-regular	P_n: $\|E\| = n-1$, ends deg 1
$K_{m,n}$: $\|E\| = m \times n$	Bipartite ⟺ no odd cycle
Adj. matrix: $O(V^2)$ space, O(1) lookup	Adj. list: O(V+E) space
Havel-Hakimi: subtract & re-sort	Odd-degree vertices: always even count

Top 12 one-liners for MCQs

1. G = (V, E): V vertices, E edges
2. Simple = no loop, no multi-edge
3. Handshaking: $\Sigma\deg(v) = 2|E|$
7. C_n is 2-regular with n edges
8. $K_{m,n}$ has m×n edges
9. Bipartite ⟺ no odd cycle

4. Digraph: $\Sigma \deg^+ = \Sigma \deg^- = |E|$

5. Walk ⊃ Trail ⊃ Path ⊃ Cycle

6. K_n has $n(n-1)/2$ edges

10. Adj. matrix $O(V^2)$; list $O(V+E)$

11. Odd-degree vertices: even count

12. Same degree seq ≠ isomorphic

Top traps: "path" in graph theory = no repeated vertex (not just any route) · cycle needs length ≥ 3 in simple graphs · Handshaking holds for multigraphs too · K_n is $(n-1)$-regular, not n-regular · adjacency matrix not symmetric for digraphs · same degree sequence ≠ isomorphic · C_n requires $n \geq 3$ · K_n bipartite only for $n \leq 2$

All tricks together: "Simple = clean" · "Every edge has two hands to shake" · "K gets all connections, C makes a ring, P draws a line" · "Even cycle = bipartite; odd cycle = not" · "Matrix is square, list is lean" · "No vertex repeated = path; no edge repeated = trail; both allowed = walk; closed path = cycle".

✓ Practice MCQs (exam pattern, 10 marks)

1. Maximum edges in a simple graph with 10 vertices:

(a) 90 (b) 45

(c) 100 (d) 50

2. An undirected graph has 12 edges. Sum of all vertex degrees:

(a) 12 (b) 6

(c) 24 (d) 144

3. In any undirected graph, the number of odd-degree vertices is:

(a) always even (b) always odd

(c) always zero (d) equal to |E|

4. In a digraph, the sum of all in-degrees equals:

(a) 2|E| (b) |V|

(c) |E| (d) $|E|^2$

5. A walk with no repeated edge but vertices may repeat is called a:

(a) trail (b) path

(c) cycle (d) bridge

6. Which sequence CANNOT be a simple graph degree sequence?

(a) (3,3,3,3) (b) (2,2,2,2)

(c) (3,3,1,1) (d) (2,2,1,1)

7. A 4-regular simple graph with 9 vertices has how many edges?

(a) 18 (b) 36

(c) 13 (d) 9

8. How many edges does K(4,5) have?

(a) 9 (b) 20

(c) 36 (d) 10

9. A graph is bipartite iff it has no:

(a) even cycle (b) odd cycle

(c) vertex of degree 1 (d) bridge

10. For a simple graph with 8 vertices and 10 edges, the complement has how many edges?

(a) 10 (b) 28

(c) 38 (d) 18

Answer key → 1-b · 2-c · 3-a · 4-c · 5-a · 6-c · 7-a · 8-b · 9-b · 10-d

Written question: State and prove the Handshaking Lemma. Hence show that every undirected graph has an even number of vertices of odd degree. Illustrate with an example. 10

1 **Statement** — For any undirected graph $G = (V, E)$, $\Sigma_{v \in V} \deg(v) = 2|E|$. 2

2 **Proof** — Each edge $\{u, v\}$ has two ends. When summing degrees, that edge is counted once at u and once at v (a loop counts twice at its single vertex). So each edge contributes exactly 2 to the sum. Hence $\Sigma \deg(v) = 2|E|$. 4

3 **Corollary** — Split V into V_even (even degree) and V_odd (odd degree). Then $\Sigma_{V_odd}\ \deg(v) = 2|E| - \Sigma_{V_even}\ \deg(v)$. The RHS is a difference of two even numbers, hence even. A sum of odd numbers is even only when the number of terms is even. Hence |V_odd| is even. 3

4 **Example** — degrees 3, 3, 2, 2, 2: sum = 12 = 2|E|, so |E| = 6. Odd-degree vertices = 2 (even). ✓ 1